

Seasonal fluctuations of phlebotomine sand fly populations (Diptera: Psychodidae) in the urban area of Marrakech, Morocco

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Abstract

Phlebotomine sand flies (Diptera: Psychodidae) were collected continuously, using sticky traps, during 1 year from October 2002 to September 2003, in an urban area of Marrakech city (Morocco). A total of 3277 specimens were collected belonging to five species. *Phlebotomus (Phlebotomus) papatasi* (54.6%) is the predominant species followed by *Sergentomyia (Sergentomyia) minuta* (20%), *S. (S.) fallax* (11.3%), *P. (Paraphlebotomus) sergenti* (10.3%) and *P. (Larroussius) longicuspis* (3.8%). Data analyses showed a mono-modal annual pattern for *P. sergenti* and a bi-modal one for the other species. *P. papatasi*, the proven vector of *Leishmania major* in Morocco, was active throughout the year. This species did not diapause in this region. *P. papatasi* population peaked in June and November, which relating to the periods of risk in this area. Its preferred temperature ranged between 32 and 36 °C but no significant correlation was found between its density and the temperature. Considering the high density and long activity period of *P. papatasi*, the area of Marrakech should be regarded as a potential focus for *L. major*. This suggests the need for a continuously surveillance to prevent risk of zoonotic cutaneous leishmaniasis.

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1. Introduction

Phlebotomine sand flies (Diptera: Psychodidae) play an important part in the transmission of leishma-

niasis, diseases caused by *Leishmania* parasitic protozoans (Kinetoplastida: Trypanosomatidae).

In Morocco, leishmaniasis is a growing public health problem. According to the Moroccan Ministry of Public Health (2001), 359 cases of cutaneous leishmaniasis (CL) due to *Leishmania major*, 1669 cases of CL due to *L. tropica*, and 116 cases of visceral leishmaniasis due to *L. infantum* were reported in 2001.

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In spite of the absence of autochthonous cases of leishmaniasis in Marrakech, this city constitutes a privileged field of study considering its proximity to the province of Chichaoua; an emerging epidemic focus of CL due to *L. tropica* (Ministry of Public Health, 2001; Guernaoui et al., in press).

In addition, based on its arid climate and its position beside the zoonotic cutaneous leishmaniasis foci in the arid region along the northern edge of the Sahara desert, Rioux et al. (1986, 1997) had classified the area of Marrakech as being at risk for *L. major*.

Studying the phlebotomine vectors bionomics is an interesting tool. It makes for a better understanding of the disease transmission dynamic and it contributes to the planning for the prevention and control against leishmaniasis particularly in the areas where the risk of leishmaniasis extension is significant.

In this paper, we present the results of a phlebotomine survey in Marrakech city, from October 2002 to September 2003, with the aim to determine the activity periods of sand flies and to specify the risk period of leishmaniasis in Marrakech city.

2. Material and methods

2.1. Study area

The investigations were conducted in Akioud, an urban district in Marrakech city (31°36'N, 8°02'W, 471 m a.s.l.). This site was selected considering its localization in the downtown of Marrakech. Moreover, former works (Guernaoui, 2000; Boussaa, 2003) showed the presence, in this site, of all the sand fly species inventoried in the urban area of Marrakech.

In Marrakech, the climate is arid. The mean maximum and minimum monthly temperatures are 41.1 and 4 °C in July and in December, respectively.

2.2. Sand flies collection and identification

Forty sticky traps (A4 blank paper sheets coated with castor oil) were set in the afternoon and collected after two days continuously from 19 October 2002 to 29 September 2003. Traps were placed, in the same places, inside and around houses and stables.

The specimens caught were preserved in 70% ethanol. They were cleared in potash 20% and Marc-

André solution (chloral hydrate/acetic acid), then dehydrated and coloured with fuschin 3‰. They were mounted in Canada balsam (Abonnenc, 1972). The identification was made by examining the morphology of male genitalia, female spermathecae and pharynges.

Records of atmospheric temperature (°C) were made simultaneously by a micro-thermometer (ACR SmartButon) placed in the station.

3. Results

During 1-year-study, 3277 sand fly specimens were caught of which 69.9% were males. Five species belonging to the genera *Phlebotomus* (68.7%) and *Sergentomyia* (31.3%) were identified. *P. papatasi* was the most prevalent species (54.6%), followed by *S. minuta* (20.0%), *S. fallax* (11.3%), *P. sergenti* (10.3%) and *P. longicuspis* (3.8%). The sex ratio was in favour of males for all species, as usually for trapping by sticky traps, with the exception of *P. sergenti* (61.4% of females) (see Table 1).

Fig. 1 shows the evolution of the mean monthly density of the total fauna during the 1-year-study. It shows that phlebotomine sand flies in this arid area were active throughout the year. Their activity was strongly marked during two periods: the first one corresponds to October–November with a peak on November (62.5 specimens per m² trap) and the second one, more important, corresponds to April–May–June–July with a peak on May (70.4 specimens per m² trap).

In these periods of high activity, the maximum and minimum monthly temperature ranged between 23 and 36 °C, and 11 and 19 °C, respectively. The monthly rainfall ranged between 35 and 50 mm.

According to the evolution of the weekly density of each species (Figs. 2 and 3), two evolution models

Table 1
Sand fly species collected in Akioud and their relative abundance (%)

Species	Male	Female	Total	(%)
<i>P. papatasi</i>	1402	386	1788	54.6
<i>P. sergenti</i>	130	207	337	10.3
<i>P. longicuspis</i>	116	9	125	3.8
<i>S. fallax</i>	228	142	370	11.3
<i>S. minuta</i>	408	249	657	20
Total	2284	993	3277	100

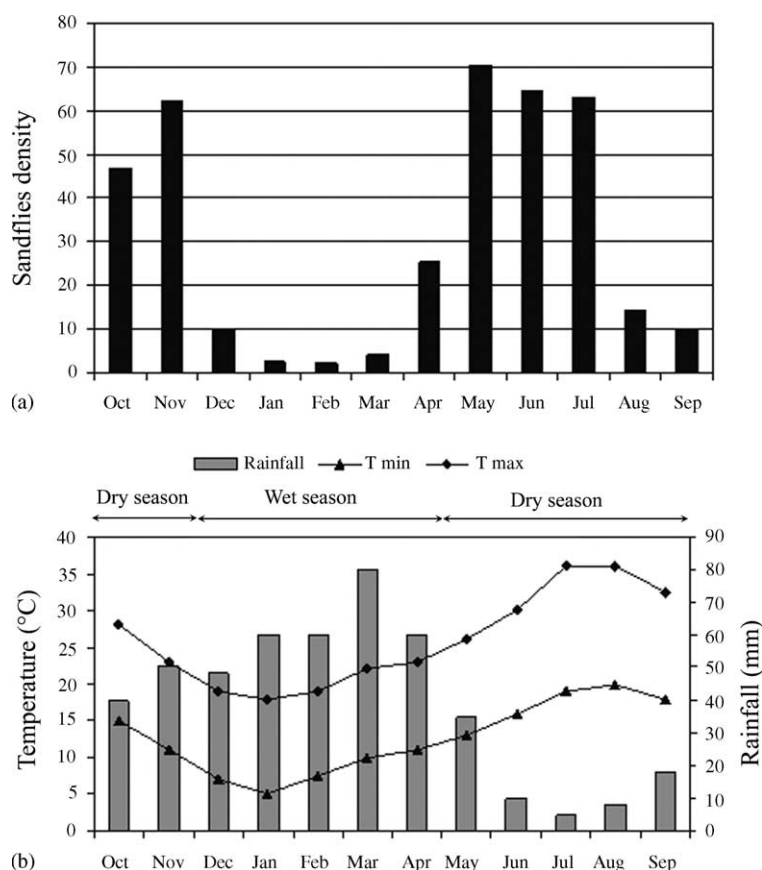


Fig. 1. (a) Mean monthly density (specimen per m² trap) of the sand fly fauna and (b) variation of the monthly temperature (maximum and minimum) and rainfall in the study area, followed from October 2002 to September 2003.

were distinguished. *P. sergenti* presented a monophasic model. It was active especially during the period April–May–June and absent in the rest of the year (Fig. 2). *P. papatasi*, *S. fallax* and *S. minuta* had already a biphasic model. *S. fallax* and *S. minuta* presented a comparable seasonality (Fig. 2). They were active during two periods, October–November and April–May–June. *S. fallax* had a more marked activity in the first period, when, *S. minuta* was more active in the second one (Fig. 2).

P. longicuspis was not abundant (3.8%). So we could not study its seasonality here. Nevertheless, our results have showed that it was active especially in November and in April–May (Fig. 2).

P. papatasi was active throughout the year. Being the most predominant species, its seasonality corresponds on that of the total fauna (Figs. 1 and 3). Two density

peaks were registered; 98 specimens per m² trap per night on 1 November and 95.8 specimens per m² trap per night on 29 June. Fig. 4 shows the mean species density of *P. papatasi* in different temperature ranges. Peaks of both sexes were recorded in the same temperature range (32–36 °C). There were no significant correlation between temperature and the species density of *P. papatasi*, but we suggest that males ($r = 0.3$) were more sensitive to temperature variations than females ($r = 0.12$).

4. Discussion

In the urban area of Marrakech city, five phlebotomine sand flies species were identified and they

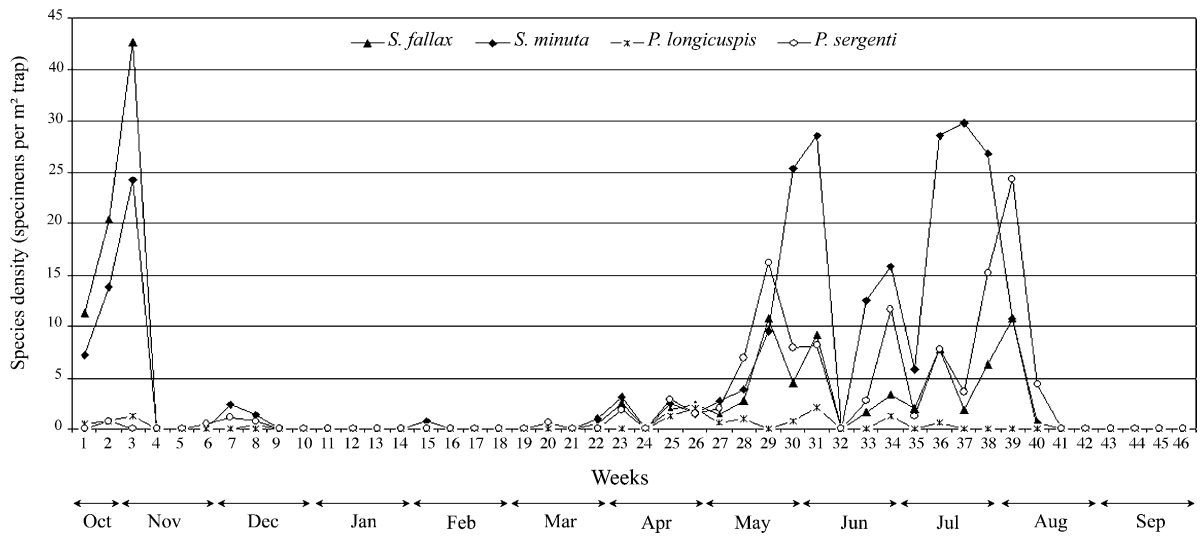


Fig. 2. Weekly density (specimen per m^2 trap) of *P. sergenti*, *P. longicuspis*, *S. fallax* and *S. minuta* in the study area followed from October 2002 to September 2003.

were abundant only in some insalubrious districts (Guernaoui, 2000; Boussaa, 2003).

P. papatasi was the predominant species, which suggests that it is well adapted to the conditions of arid climate (Rioux et al., 1984, 1997).

In this study, the seasonality of the sand fly fauna in Marrakech was examined. Sand flies were active especially during the dry season (from May to November). Two activity peaks were noted in May and November which corresponds to the wettest months during the

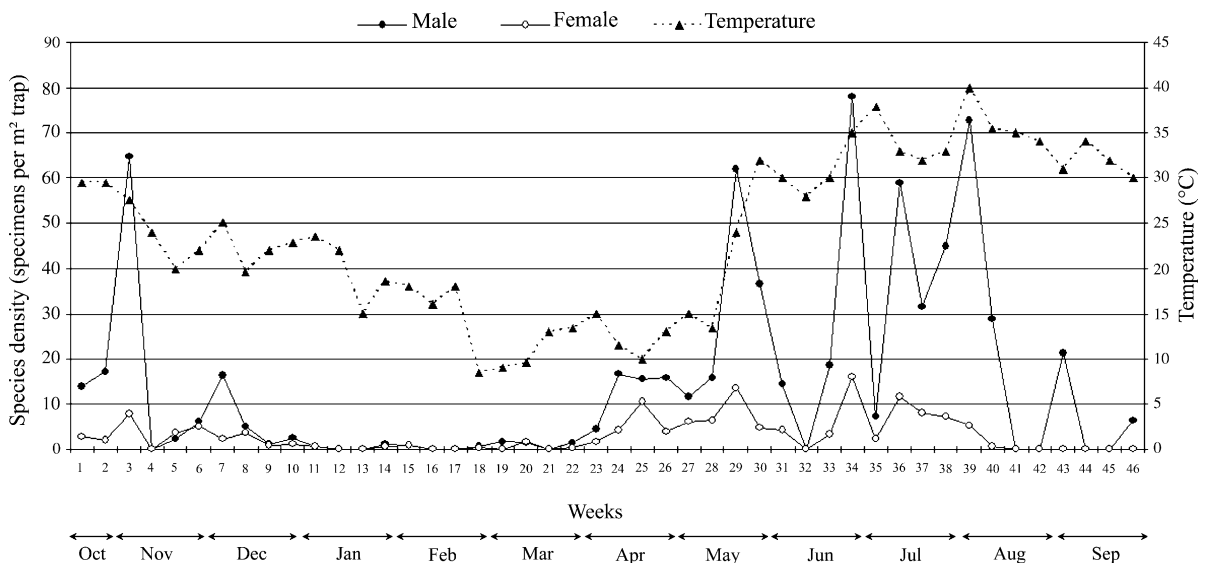


Fig. 3. Weekly density (specimen per m^2 trap) of *P. papatasi* (male and female) with variations of mean weekly temperature in the study area, followed from October 2002 to September 2003.

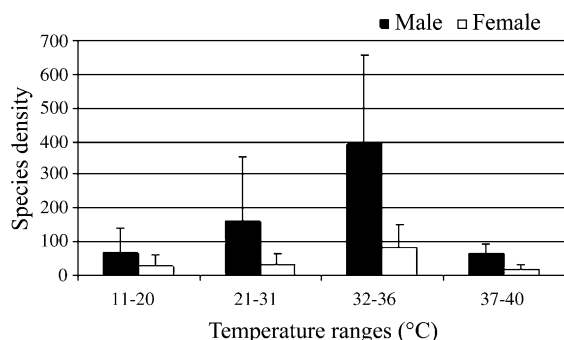


Fig. 4. Mean species density of *P. papatasi* in different temperature ranges.

dry season. The specific richness was maximal (five species) in May when the daily temperature ranged between 15 and 32 °C and rainfall was 35 mm.

The species differed in their activity periods. Two seasonal patterns were identified, a mono-modal one for *P. sergenti* and a bi-modal one for *P. papatasi*, *S. minuta* and *S. fallax*. In mountainous areas near Marrakech (Guernaoui et al., in press), the period of sand flies activity was short (from June to November) and all species presented a monophasic evolution. This could be the result of the difference in climatic factors between the plain and the mountain. In northern Israel, sand fly populations were prevalent from April to November and peaked between June and August (Kravchenko et al., 2004).

We gave a particular intention to *P. papatasi* in view of its abundance in this area and its role in transmitting *L. major* in Morocco (Rioux et al., 1986). It was active throughout the year, absent occasionally in some samples because of the meteorological disturbances. These results suggest that this vector did not diapause in this area. Two density peaks of *P. papatasi* were identified, the first one in the beginning June (summer) and the second one in the beginning November (autumn). Between these two peaks, the density presented significant microvariations due especially to the effect of the wind. These seasonal patterns should be related to different environmental factors specially climatologically ones. In Israel, Wasserberg et al. (2003) found that *P. papatasi*, in conditions of relatively mesic desert, peaked in spring (April or May), while in extremely xeric area, its population density was bi-modal with two peaks in both spring and autumn. In the arid area of India, Singh (1999) demonstrated the significant

role of the temperature on *P. papatasi* distribution and Srinivasan et al. (1993) showed a significant positive correlation between *P. papatasi* density and rainfall. In this study, the preferred temperature for *P. papatasi* ranged from 32 to 36 °C but no significant correlation was found. It was less active at temperatures of 11–20 and 37–40 °C. Moreover, Killick-Kendrick (1999) and Wasserberg et al. (2003) had noticed that low temperatures and also the hot conditions are limiting factor of its activity.

In conclusion, *P. papatasi* density was very important in Marrakech throughout the year. In addition, its position close to the zoonotic cutaneous leishmaniasis foci in the arid region along the northern edge of the Sahara desert (Rioux et al., 1986) and the omnipresence of *Meriones shawi*, the main *L. major* reservoir host in Morocco, Marrakech city should be regarded as a potential focus for *L. major*. This suggests the need for a continuously surveillance to prevent risk of zoonotic cutaneous leishmaniasis. *P. papatasi* population peaked in June and November, which correspond to the periods of risk in this area.

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